

Detailed Design Report (DDR)

Project: YB Vehicle Rental Tracking System

2.1 Database

2.1.1 Relational Database

The figure below shows the relation between the tables in the YB Vehicle Rental Tracking System database.

```
erDiagram
    Branches ||--o{ Vehicles : "CurrentBranchID"
    Branches ||--o{ Employees : "BranchID"
    Branches ||--o{ Rentals : "PickUpBranchID"
    Branches ||--o{ Rentals : "DropOffBranchID"
    Vehicle_Categories ||--o{ Vehicle_Models : "CategoryID"
    Vehicle_Models ||--o{ Vehicles : "ModelID"
    Vehicles ||--o{ Rentals : "VehicleID"
    Vehicles ||--o{ Maintenance_Logs : "VehicleID"
    Customers ||--o{ Rentals : "CustomerID"
    Employees ||--o{ Rentals : "EmployeeID"
    Employees ||--o{ Employees : "SupervisorID"
    Rentals ||--o{ Payments : "RentalID"
    Rentals ||--o{ Damage_Reports : "RentalID"
```

2.1.2 Database Schema / Data Dictionary

Branches Table

Column_name	Type	Nullable	Primary Key
BranchID	INTEGER	NO	YES
BranchName	NVARCHAR(100)	NO	NO
StreetAddress	NVARCHAR(150)	NO	NO
Barangay	NVARCHAR(50)	NO	NO
City	NVARCHAR(50)	NO	NO
Province	NVARCHAR(50)	NO	NO
Phone	VARCHAR(20)	NO	NO

Vehicle_Categories Table

Column_name	Type	Nullable	Primary Key
CategoryID	INTEGER	NO	YES
CategoryName	NVARCHAR(50)	NO	NO
DailyRate	DECIMAL(10,2)	NO	NO
OverdueRatePerHour	DECIMAL(10,2)	NO	NO

Vehicle_Models Table

Column_name	Type	Nullable	Primary Key
ModelID	INTEGER	NO	YES
Brand	NVARCHAR(50)	NO	NO
ModelName	NVARCHAR(100)	NO	NO
FuelType	VARCHAR(20)	NO	NO
Transmission	VARCHAR(20)	NO	NO
CategoryID	INT	NO	NO

Vehicles Table

Column_name	Type	Nullable	Primary Key
VehicleID	INTEGER	NO	YES
ModelID	INT	NO	NO
LicensePlate	VARCHAR(20)	NO	NO
CurrentMileage	INT	NO	NO
Status	VARCHAR(20)	NO	NO
CurrentBranchID	INT	NO	NO

Customers Table

Column_name	Type	Nullable	Primary Key
CustomerID	INTEGER	NO	YES
FirstName	NVARCHAR(50)	NO	NO
LastName	NVARCHAR(50)	NO	NO
Email	VARCHAR(100)	NO	NO
Phone	VARCHAR(20)	NO	NO
DriversLicense	VARCHAR(30)	NO	NO
LicenseExpiry	DATE	NO	NO
IsActive	BIT	NO	NO

Employees Table

Column_name	Type	Nullable	Primary Key
EmployeeID	INTEGER	NO	YES
FirstName	NVARCHAR(50)	NO	NO

Column_name	Type	Nullable	Primary Key
LastName	NVARCHAR(50)	NO	NO
Email	VARCHAR(100)	NO	NO
Role	NVARCHAR(50)	NO	NO
BranchID	INT	NO	NO
SupervisorID	INT	YES	NO

Rentals Table

Column_name	Type	Nullable	Primary Key
RentalID	INTEGER	NO	YES
EmployeeID	INT	NO	NO
CustomerID	INT	NO	NO
VehicleID	INT	NO	NO
PickUpBranchID	INT	NO	NO
DropOffBranchID	INT	YES	NO
RentedOn	DATETIME	NO	NO
ExpectedReturn	DATETIME	NO	NO
ActualReturn	DATETIME	YES	NO
StartMileage	INT	NO	NO
EndMileage	INT	YES	NO
Status	VARCHAR(20)	NO	NO

Payments Table

Column_name	Type	Nullable	Primary Key
PaymentID	INTEGER	NO	YES
RentalID	INT	NO	NO
PaidOn	DATETIME	NO	NO
BaseAmount	DECIMAL(10,2)	NO	NO
PenaltyAmount	DECIMAL(10,2)	NO	NO
TotalAmount	DECIMAL(10,2)	NO	NO
PaymentMethod	VARCHAR(30)	NO	NO

Maintenance_Logs Table

Column_name	Type	Nullable	Primary Key
LogID	INTEGER	NO	YES

Column_name	Type	Nullable	Primary Key
VehicleID	INT	NO	NO
StartDate	DATETIME	NO	NO
EndDate	DATETIME	YES	NO
Cost	DECIMAL(10,2)	YES	NO
Description	TEXT	NO	NO
Status	VARCHAR(20)	NO	NO

Damage_Reports Table

Column_name	Type	Nullable	Primary Key
ReportID	INTEGER	NO	YES
RentalID	INT	NO	NO
IncidentDate	DATETIME	NO	NO
Description	TEXT	NO	NO
EstimatedRepairCost	DECIMAL(10,2)	NO	NO
Status	VARCHAR(20)	NO	NO

2.2 Input Forms

2.2.1 Input Inventory

Form Name	Description
Splash/Landing Screen	Allows the user to select the Target Database environment (e.g. YB_Rentals_DB vs Test DB) before logging in.
Navigation Sidebar	Contains buttons mapped to all 10 major database entities for easy switching.
CRUD Panel	Contains textboxes, dropdowns, and date inputs specific to the active table. Includes "Add New", "Save Details", "Delete Selected", and "Clear Form" actions.
Dynamic Filter Panel	Allows users to build complex WHERE clause searches dynamically using logical operators (And/Or) and arithmetic operators (>, <, IS, etc.). Filters can be saved/loaded to a local .json file.

2.2.2 Input Forms Data Dictionary

(Note: Represents the Tkinter GUI inputs mapped to the database)

Form	Actions Permitted	Allow Nulls
Branches (Name, Street, Barangay, City, Province, Phone)	ADD, EDIT, DELETE	NO
Vehicle Categories (Name, Daily Rate, Overdue Rate)	ADD, EDIT, DELETE	NO
Vehicle Models (Brand, Model, Fuel, Trans, Category)	ADD, EDIT, DELETE	NO
Vehicles (Model, License Plate, Mileage, Status, Branch)	ADD, EDIT, DELETE	NO
Customers (Name, Email, Phone, License, Is Active)	ADD, EDIT, DELETE	NO
Employees (Name, Email, Role, Branch, Supervisor)	ADD, EDIT, DELETE	YES (Supervisor)
Rentals (Emp, Cust, Vehicle, Pick/Drop Branch, Dates, Mileage, Status)	ADD, EDIT, DELETE	YES (Drop Branch, End Mileage, Actual Return)
Payments (Rental, Date, Base/Penalty/Total Amt, Method)	ADD, EDIT, DELETE	NO
Maintenance Logs (Vehicle, Dates, Cost, Desc, Status)	ADD, EDIT, DELETE	YES (End Date, Cost)
Damage Reports (Rental, Date, Desc, Est Cost, Status)	ADD, EDIT, DELETE	NO

2.3 Screens

2.3.1 Screen Navigation Diagram

```
flowchart TD
    Splash["Splash[Landing Screen<br/>Database Selector]"] --> Main["Main[Main Application GUI]"]
    Main --> Branches["Branches[Branches View]"]
    Main --> Categories["Categories[Categories View]"]
    Main --> Models["Models[Models View]"]
    Main --> Vehicles["Vehicles[Vehicles View]"]
    Main --> Customers["Customers[Customers View]"]
    Main --> Rentals["Rentals[Rentals View]"]
    Main --> Payments["Payments[Payments View]"]
    Main --> Maintenance["Maintenance[Maintenance View]"]
    Main --> Damage["Damage[Damage Reports View]"]
    Main --> Employees["Employees[Employees View]"]
```

2.3.2 Screen Description/Layout

Screen	Description
Landing Screen	Initial prompt allowing the administrator to route the application to the Production Database or the Testing Database.
Main GUI Window	The core application layout containing the Sidebar (Navigation), the Top Bar (Dynamic Filters), and the Center Panel (Data Grid and Input Form).
Dynamic Filter Container	A sub-section appearing above the Data Grid. It lets users add/remove condition rows, rearrange their priority (▲/▼), and save/load filter configurations.
Input Form Container	The sub-section below the Data Grid. Its text boxes and labels dynamically morph to match the schema of the currently selected table in the sidebar.